

Software Test Plan (STP) - Blood Bank Management System (BBMS)

Project: Blood Bank Management System (BBMS) - Software Engineering Mini-Project

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Technology: Python (Django) or JavaScript (Node.js/Express), HTML/CSS/JavaScript, SQLite

1. Introduction

Purpose: This document defines the test plan for the Blood Bank Management System (BBMS) v1.0. It outlines objectives, scope, strategy, resources, schedule, and responsibilities for testing, and is aligned with the functional, non-functional, and security requirements of the BBMS SRS v1.0. It provides a repeatable method for verifying that implemented BBMS behavior satisfies the specified requirements.

Test objectives:

- Verify that each major functional workflow satisfies its corresponding BBMS-F requirement.
- Verify that measurable non-functional requirements are validated using defined thresholds and test methods.
- Validate that security controls satisfy BBMS-SR-001 through BBMS-SR-008.
- Identify defects, inconsistencies, boundary conditions, and integration failures before acceptance.
- Maintain bidirectional traceability between SRS requirements and test cases.

Scope: Testing covers the browser-based BBMS workflows from donor and hospital registration through donation, testing, inventory, requisition, compatibility checking, allocation, dispatch, delivery confirmation, auditing, administration, reporting, and public availability. Non-functional validation covers performance, reliability, usability/accessibility, maintainability, portability, and scalability. Security validation covers authentication, password protection, server-side RBAC, SQL injection prevention, input validation/XSS, session security, brute-force/rate limiting, HTTPS/TLS, and auditability.

References: BBMS SRS v1.0; BBMS Software Architecture & Design Specification (SAD) v1.0; IEEE Std 830-1998 (structural reference used by the SRS); course guidance for Software Engineering Mini-Project Part 1.

Definitions:

Term	Meaning
BBMS	Blood Bank Management System

Term	Meaning
SRS	Software Requirements Specification
STP	Software Test Plan
RTM	Requirements Traceability Matrix
FEFO	First-Expiry-First-Out
RBAC	Role-Based Access Control
API / REST	Application Programming Interface / Representational State Transfer
SQLi	SQL Injection
XSS	Cross-Site Scripting
STAT	Immediate / emergency priority request
Bag ID	Unique identifier assigned to a collected blood unit
WCAG	Web Content Accessibility Guidelines

2. Test Items

BBMS is a standalone academic web application supporting the workflow donor → collection → testing → inventory → requisition → dispatch. It uses a browser-based client, a Python (Django) or Node.js (Express) backend, an ORM/data-access layer, and an embedded SQLite database. Outbound email notifications use SMTP. The following modules are under test:

Module	Primary scope	SRS coverage
Authentication / Access	Registration, login, reset, sessions, RBAC	F-001–F-005; F-040–F-042; F-060–F-063; SR-001–SR-007
Donor	Profiles, eligibility, appointments, history	F-006–F-019
Collection / Blood Unit	Collection, Bag ID, blood group	F-017–F-023
Testing / Quarantine	Serology, quarantine, discard	F-024–F-030
Inventory / FEFO	Components, counts, expiry, FEFO	F-031–F-039
Hospital / Requisition	Registration, requisitions, STAT queue	F-040–F-048
Matching / Allocation	Compatibility and allocation	F-049–F-055
Dispatch / Delivery	Dispatch records and receipt	F-052–F-057
Audit / Admin	Audit, users, roles	F-058–F-063; SR-008
Reports / Public Portal	Reports and availability lookup	F-064–F-068
Quality attributes	Performance, reliability, UX, portability, scale	NF-001–NF-015

User roles exercised in testing: Donor; Hospital User; Blood Bank Staff; Administrator; Public / Guest (public blood availability only).

3. Features to be Tested

Features mapped to SRS requirement IDs:

Feature	SRS requirement IDs
Donor registration, age validation, unique email, authentication, password reset	F-001–F-005
Donor profile, donation history, next eligible donation date	F-006–F-008
Medical eligibility questionnaire and outcome recording	F-009–F-011
Appointment booking, slot capacity, reschedule, cancel, notification	F-012–F-016
Blood collection, volume/time capture, donor history update	F-017–F-019
Unique Bag ID generation and duplicate prevention	F-020–F-021
ABO/Rh blood-group support and confirmed blood-group correction	F-022–F-023
Serology tests, quarantine default, pass/fail, discard on reactive result	F-024–F-027
Quarantine visibility, availability exclusion, status audit, quarantine list	F-028–F-030
Component separation and origin traceability	F-031–F-032
Real-time inventory, filtering, count updates	F-033–F-036
Expiry calculation, FEFO ordering, automatic expiry handling	F-037–F-039
Hospital registration/approval/login	F-040–F-042
Hospital requisitions, IDs, approval/partial/rejection, history	F-043–F-046
Emergency/STAT prioritization and alerts	F-047–F-048
Compatibility determination, allocation restrictions/override, outcome recording	F-049–F-051
Allocation, dispatch records, status updates, fulfilment status	F-052–F-055
Delivery confirmation and delivered status	F-056–F-057
Audit logging and administrator-only read-only audit access	F-058–F-059
Admin user/role management and hospital approval	F-060–F-063
CSV reports and public blood availability without identifying data	F-064–F-068
Performance, reliability, usability/accessibility, maintainability, portability, scalability	NF-001–NF-015
Authentication, password hashing, RBAC, SQLi prevention, XSS, session security, brute-force/rate limiting, HTTPS/TLS, auditability	SR-001–SR-008

4. Features Not to be Tested

The following are out of scope because the SRS identifies them as excluded from the mini-project. They may be reviewed for scope compliance but are not functional test targets.

- Integration with national or regional blood-bank registries or government health systems (SRS 1.2).
- Payment processing or billing of any kind (SRS 1.2).
- Direct interfacing with physical laboratory testing equipment; serology results are entered manually by staff (SRS 1.2 and 2.6).
- Native mobile applications; testing is limited to the responsive web application (SRS 1.2).
- Multi-language localization (SRS 1.2).

5. Test Approach / Strategy

Testing uses a risk-based, requirement-driven approach. Functional tests use representative positive, negative, boundary, and authorization cases. Non-functional tests use the measurable SRS thresholds. Security validation is performed before acceptance of security-sensitive functionality.

Levels:

- **Unit tests** (module-level): business-logic modules and validation routines.
- **Integration tests:** frontend → REST API, API → SQLite/ORM, authentication/authorization middleware, collection → inventory, testing → quarantine, requisitions → matching, allocation → dispatch, notifications → SMTP, and audit logging on all data-changing operations.
- **System tests** (end-to-end): donor registration → eligibility → appointment → collection → testing → availability; and hospital registration → requisition → approval → matching → allocation → dispatch → delivery confirmation.
- **Acceptance tests:** course-evaluator review of execution evidence and the completed RTM against the SRS acceptance criteria.

Types:

- **Functional testing:** each workflow verified against its SRS acceptance condition, including valid and invalid input, boundary cases, duplicate data, unavailable inventory, incompatible blood groups, unauthorized roles, and state transitions.
- **Regression testing:** rerun after changes affecting authentication, inventory, blood status transitions, requisition allocation, cross-matching, reporting, or database schemas; previously passed tests are executed again against affected modules.
- **Performance testing:** blood-availability search within 2 seconds under up to 50 concurrent users (NF-001); appointment booking within 3 seconds (NF-002); at least 50 concurrent authenticated users without exceeding the stated response limits (NF-003).
- **Reliability / stress testing:** 99% scheduled availability (NF-004); committed transactions persist across application restart (NF-005); unhandled errors logged server-side with timestamp and stack trace (NF-006).
- **Usability / accessibility testing:** first-time donor completes registration unaided in under 5 minutes (NF-007); primary workflows meet WCAG 2.1 Level AA colour-contrast and keyboard-navigation requirements (NF-008); validation errors identify the affected field and issue (NF-009).
- **Portability / compatibility testing:** latest two stable versions of Chrome, Firefox, and Edge (NF-012); deployment on a host supporting Python 3.10+ or Node.js 18+ without code modification (NF-013).
- **Maintainability and scalability review:** NF-010, NF-011, NF-014, and NF-015 validated through code review, unit-test coverage, schema review, and architecture review as applicable.

Entry Criteria:

- Approved SRS and Test Plan baseline is available.

- Build/deployment candidate is available in a controlled test environment.
- Database schema and required seed/configuration data are available.
- Test accounts for Donor, Hospital User, Blood Bank Staff, and Administrator exist or can be created.
- Required test tools and browser versions are available.
- No known blocker prevents execution of core test workflows.

Exit Criteria:

- All planned functional and security test cases have been executed or formally marked Blocked with reason.
- No unresolved Critical or High severity defect remains for an acceptance-critical feature.
- All mandatory performance thresholds in NF-001 through NF-003 have been evaluated.
- Reliability, usability/accessibility, portability, maintainability, and scalability requirements have been assessed using their defined methods.
- The RTM is updated with actual test status and defect references.

Pass / Fail Criteria: A test case is **Pass** when all stated expected results are observed without an unacceptable defect. It is **Fail** when any mandatory expected result is not satisfied. It is **Blocked** when execution cannot proceed because of an environment, dependency, or unresolved prerequisite issue. **Not Run** is used before execution. No result is pre-labelled Pass in this plan.

5.1 Security Validation

Security testing validates each SRS security requirement using controlled positive and negative test cases, performed against a non-production test environment with no real donor or hospital data.

Test case	Req.	Area	Method	Expected security result
TC-SEC-01	SR-001	Authentication	Attempt access to a role-specific function without login; verify authenticated access succeeds.	Unauthenticated request is denied; authenticated valid user is granted access appropriate to role.
TC-SEC-02	SR-002	Password hashing	Register/change password; inspect stored credential representation and logs/responses.	Only one-way salted hashes are stored; plaintext password is absent from logs/responses.
TC-SEC-03	SR-003	RBAC	Attempt donor/hospital access to staff/admin endpoint; verify server-side authorization.	Unauthorized request receives HTTP 403; protected data/action is not exposed.
TC-SEC-04	SR-004	SQL injection	Submit SQL injection payloads in authentication/search/form inputs; review API/database behavior.	Input is treated as data; no unauthorized query execution or data exposure.
TC-SEC-05	SR-005	Input validation / XSS	Submit malformed fields and stored/reflected XSS probes.	Invalid payloads are rejected/encoded; script content is not executed.
TC-SEC-06	SR-006	Session / cookies	Inspect session cookies; test logout and inactivity expiration.	Cookies are HttpOnly and Secure with defined expiry; logout and 30-minute inactivity invalidate the session.

Test case	Req.	Area	Method	Expected security result
TC-SEC-07	SR-007	Brute-force / rate limit	Perform repeated failed logins and rapid login requests.	Account locks for 15 minutes after 5 consecutive failures; requests beyond 10/IP/min are rate-limited.
TC-SEC-08	SR-008	HTTPS/TLS + audit	Connect over HTTP/HTTPS and perform security-relevant actions.	Traffic uses HTTPS/TLS; login attempts, lockouts, role changes and other security events are recorded in a tamper-evident audit trail.

6. Test Environment

Category	Planned environment
Hardware	Standard PC/laptop or single virtual machine; no specialized hardware required.
Server OS	Host OS supporting Python 3.10+ or Node.js 18+.
Software (backend)	Python 3.10+ with Django, or Node.js 18+ with Express, per implementation choice.
Database	SQLite embedded database via ORM/data-access layer.
Client	Modern JavaScript-enabled browser over LAN/internet connectivity.
Browsers	Latest two stable versions of Chrome, Firefox, and Edge for portability testing.
Network	HTTPS/TLS for client-server communication; SMTP for outbound email tests.
Tools	Node.js test runner and Supertest where applicable (API/unit); ApacheBench (load); Lighthouse (accessibility); controlled SQL injection testing tool / manual payloads.
Test data isolation	Synthetic donor, hospital, inventory, and requisition records only; no production/real patient data.

Tool usage is planned from the SRS-described test approach; this document does not claim that any test has already been executed.

Test Data

Data category	Examples / conditions
Donors	Valid adult donor; under-18 donor; duplicate email; valid and invalid contact details.
Eligibility	All eligible answers; each configured disqualifying condition; recorded outcome linked to appointment.
Appointments	Available slot; full-capacity slot; cancellation; reschedule inside/outside 24-hour window.
Blood units	Unique Bag IDs; duplicate ID attempt; all eight ABO/Rh groups; component types.
Serology	Pass for all required tests; single Fail; incomplete test set; quarantine transitions.
Inventory	Available, Quarantined, Discarded, Expired, Dispatched, Delivered units; multiple expiry dates for FEFO.
Hospitals	Pending approval; approved; rejected; deactivated credentials.
Requisitions	Routine and Emergency/STAT; full, partial, rejected, incompatible and compatible requests.
Security inputs	SQL injection strings; XSS probes; invalid payload sizes/types; repeated failed logins.
Performance data	Synthetic concurrent sessions and repeated blood-availability queries up to 50 users.

7. Test Schedule

Milestone	Planned activity	Target
T1	Test-plan baseline and environment readiness	Before system test execution
T2	Functional and integration test execution	After build is available
T3	Security validation	After authentication/RBAC/input controls are integrated
T4	Performance, reliability, accessibility, portability tests	After core workflows stabilize
T5	Regression and retesting	After defect fixes
T6	Final traceability and test summary	Before project acceptance

8. Test Deliverables

- Approved Software Test Plan (.docx).
- Detailed functional and non-functional test cases.
- Security validation test set.
- Test execution records with Actual Result and Status fields.
- Defect records and retest evidence, if defects are found.
- Updated Requirements Traceability Matrix.
- Test summary / acceptance report after execution.

9. Roles and Responsibilities

Team T9: Uttam (PES1UG24CS697), Abhinav K (PES1UG24CS701), Akshay Arcot (PES1UG24CS705), Navaneeth Tanuboddi (PES1UG24CS709). Role-to-member assignment is to be confirmed.

Role	Name	Responsibility
Project / Test Lead	To be assigned	Maintain the Test Plan, coordinate execution, review evidence, update traceability and test summary.
Tester	To be assigned	Execute test cases, record evidence, log and classify defects, perform retesting/regression.
Developer / Module Owner	To be assigned	Provide builds, support test-data setup, fix defects, and support root-cause analysis.
Security Validation Owner	To be assigned	Execute SR-001-SR-008 validation (TC-SEC-01 to TC-SEC-08) and document security findings.
Academic Evaluator	Course faculty / evaluator	Review deliverables and acceptance evidence as required by the course.

10. Risks and Mitigation

Risk	Impact	Mitigation / contingency
SQLite write-concurrency limitations	Performance/availability limits during concurrent tests	Keep load within SRS mini-project scope; measure and document observed limits.
Unstable build or incomplete integration	Blocks functional tests	Freeze a test build before execution; maintain a known-good baseline.

Risk	Impact	Mitigation / contingency
Incorrect test data or time configuration	False failures in expiry/eligibility tests	Use seeded, versioned test data and server-authoritative timestamps.
Security test payloads affect test data	Data contamination	Use an isolated test database and synthetic accounts only.
Email/SMTP unavailable	Notification tests blocked	Record dependency as Blocked; verify non-email on-screen behavior separately where specified.
Browser/version differences	UI compatibility defects	Execute portability tests on required browsers and record deviations.
High-severity defect discovered late	Acceptance delay	Prioritize blocker/high-severity fixes and focused regression on affected modules.

11. Assumptions & Dependencies

- The SRS v1.0 and SAD v1.0 baselines (requirement IDs and the API contract) remain stable during test execution.
- Synthetic test data and test accounts for all four roles will be provided before execution.
- An HTTPS/TLS-enabled, non-production test environment is available (required for TC-SEC-06 and TC-SEC-08).
- An SMTP service or test mail sink is available for notification tests; otherwise dependent tests are marked Blocked.
- The server clock or test configuration can be advanced to exercise expiry, eligibility-interval, and 30-minute inactivity behavior.
- Load stays within the SRS mini-project scale (up to 50 concurrent users) on a single-host SQLite deployment.
- Required browsers (latest two stable Chrome, Firefox, Edge) and tools (Supertest, ApacheBench, Lighthouse) can be installed.

12. Suspension & Resumption Criteria

Suspend testing if:

- The build cannot start or the database is unavailable.
- A blocker prevents execution of core workflows.
- Test data is corrupted.
- The environment is materially different from the approved baseline.

Resume testing if:

- The blocker is resolved.
- The environment is revalidated.
- Affected tests are rerun from the applicable baseline.

13. Test Case Management & Traceability

The RTM maps every SRS requirement group to planned test cases. Status is **Not Run** until execution occurs. Where one test case validates a group of closely related requirements, the mapping is explicit.

SRS Req.	Module / feature	Test case(s)	Verification type	Status
F-001–F-005	Donor registration/login/recovery	TC-AUTH-01, TC-AUTH-02	Functional / Security	Not Run
F-006–F-008	Donor profile/history/eligibility date	TC-DNR-01	Functional	Not Run
F-009–F-011	Medical eligibility questionnaire	TC-ELG-01	Functional	Not Run
F-012–F-016	Appointment booking/reschedule/cancel/notification	TC-APT-01, TC-APT-02	Functional / Integration	Not Run
F-017–F-019	Blood collection	TC-COL-01	Functional	Not Run
F-020–F-021	Bag ID uniqueness	TC-BAG-01	Functional	Not Run
F-022–F-023	Blood group management	TC-BAG-02	Functional	Not Run
F-024–F-027	Serology and unsafe-unit handling	TC-SER-01, TC-SER-02	Functional / Security	Not Run
F-028–F-030	Quarantine controls	TC-QNT-01	Functional / Audit	Not Run
F-031–F-032	Component separation	TC-CMP-01	Functional	Not Run
F-033–F-036	Inventory management	TC-INV-01, TC-INV-02	Functional / Integration	Not Run
F-037–F-039	Expiry and FEFO	TC-EXP-01	Functional	Not Run
F-040–F-042	Hospital registration/login	TC-AUTH-03	Functional / Authorization	Not Run
F-043–F-046	Blood requisition	TC-REQ-01, TC-REQ-02	Functional	Not Run
F-047–F-048	Emergency/STAT	TC-REQ-03	Functional / Integration	Not Run
F-049–F-051	Compatibility/cross-matching	TC-MATCH-01	Functional	Not Run
F-052–F-055	Allocation/dispatch	TC-DISP-01	Functional / Integration	Not Run
F-056–F-057	Delivery confirmation	TC-DEL-01	Functional	Not Run
F-058–F-059	Audit logging	TC-AUD-01	Functional / Security	Not Run
F-060–F-063	Admin/user/role management	TC-ADM-01	Functional / Security	Not Run
F-064–F-066	Reports/analytics	TC-RPT-01	Functional	Not Run
F-067–F-068	Public availability/privacy	TC-PUB-01	Functional / Security	Not Run
NF-001–NF-003	Performance/concurrency	TC-NF-01	Performance	Not Run
NF-004–NF-006	Reliability/availability/error logging	TC-NF-02	Reliability	Not Run
NF-007–NF-009	Usability/accessibility/errors	TC-NF-03	Usability / Accessibility	Not Run
NF-010–NF-011	Maintainability/unit-test coverage	TC-NF-04	Maintainability	Not Run
NF-012–NF-013	Portability/browser/runtime	TC-NF-05	Portability	Not Run
NF-014–NF-015	Scalability/location extension	TC-NF-06	Scalability / Architecture	Not Run

SRS Req.	Module / feature	Test case(s)	Verification type	Status
SR-001	Authentication	TC-SEC-01	Security	Not Run
SR-002	Password hashing	TC-SEC-02	Security	Not Run
SR-003	RBAC/authorization	TC-SEC-03	Security	Not Run
SR-004	SQL injection prevention	TC-SEC-04	Security	Not Run
SR-005	Input validation/XSS	TC-SEC-05	Security	Not Run
SR-006	Session/cookie security	TC-SEC-06	Security	Not Run
SR-007	Brute-force/rate limiting	TC-SEC-07	Security	Not Run
SR-008	HTTPS/TLS and security auditability	TC-SEC-08, TC-AUD-01	Security / Audit	Not Run

13.1 Detailed Test Cases

TC-AUTH-01 — Valid donor registration and login

Requirement ID	BBMS-F-001, F-002, F-003, F-004
Module / Feature	Authentication
Priority	High
Objective	Verify valid donor registration, age validation, unique email, and subsequent authentication.
Preconditions	Registration page available; email not previously registered.
Test Data	Adult donor details; unique email; valid password.
Test Steps	1. Submit valid registration. 2. Confirm account creation. 3. Log in with the registered credentials. 4. Attempt duplicate registration with the same email.
Expected Result	Valid donor account is created and can authenticate; under-age data is rejected; duplicate email registration is rejected.
Actual Result	[To be recorded during execution]
Status	Not Run

TC-AUTH-02 — Password reset and logout lifecycle

Requirement ID	BBMS-F-005
Module / Feature	Authentication / Recovery
Priority	Medium
Objective	Verify the time-limited single-use password reset link and session logout behavior.
Preconditions	Registered user with access to the registered email.
Test Data	Synthetic account, valid reset request, expired/reused reset link.
Test Steps	1. Request password reset. 2. Use reset link once. 3. Attempt to reuse it. 4. Log in with the new password. 5. Log out and access a protected page.
Expected Result	Valid reset link changes the password once; reuse is rejected/invalid; logout prevents further protected access.
Actual Result	[To be recorded during execution]
Status	Not Run

TC-DNR-01 — Donor profile and history

Requirement ID	BBMS-F-006–F-008
Module / Feature	Donor Profile
Priority	Medium
Objective	Verify profile update, donation history display, and next eligible donation date calculation.
Preconditions	Authenticated donor with prior successful donation record.
Test Data	Updated contact/address/blood group; prior donation date.
Test Steps	1. Open profile. 2. Update permitted fields. 3. View donation history. 4. Compare displayed next eligible date to configured interval.
Expected Result	Profile changes persist; history shows date/location/Bag ID; next eligible date reflects the configured interval.
Actual Result	[To be recorded during execution]
Status	Not Run

TC-ELG-01 — Medical eligibility questionnaire

Requirement ID	BBMS-F-009–F-011
Module / Feature	Eligibility
Priority	High
Objective	Verify questionnaire is completed before booking and disqualifying answers produce temporary ineligibility.
Preconditions	Donor has a selectable appointment slot.
Test Data	One eligible response set and one disqualifying response set.
Test Steps	1. Attempt booking without questionnaire. 2. Complete eligible questionnaire and book. 3. Repeat with disqualifying response.
Expected Result	Booking cannot be confirmed before questionnaire; eligible donor can proceed; disqualifying answer marks donor temporarily ineligible and records outcome.
Actual Result	[To be recorded during execution]
Status	Not Run

TC-APT-01 — Appointment capacity and cancellation

Requirement ID	BBMS-F-012, F-013, F-015, F-016
Module / Feature	Appointment Booking
Priority	High
Objective	Verify slot availability/capacity, cancellation, and notification behavior.
Preconditions	A configured appointment slot with known capacity.
Test Data	Bookings up to capacity; cancellation request.
Test Steps	1. Fill slot to capacity. 2. Attempt an additional booking. 3. Cancel an existing appointment. 4. Retry booking.
Expected Result	Capacity limit prevents excess booking; cancellation releases capacity; confirmation is shown and/or sent according to the SRS.
Actual Result	[To be recorded during execution]
Status	Not Run

TC-APT-02 — Appointment rescheduling boundary

Requirement ID	BBMS-F-014
Module / Feature	Appointment Rescheduling
Priority	Medium
Objective	Verify rescheduling is allowed up to 24 hours before the appointment and blocked inside the 24-hour window.
Preconditions	Existing donor appointment and available alternative slots.
Test Data	Appointment at exactly 24 hours, and another inside 24 hours.
Test Steps	1. Attempt reschedule at 24-hour boundary. 2. Attempt reschedule within 24 hours. 3. Review appointment record.
Expected Result	Boundary/allowed case succeeds according to configured rule; inside-24-hour case is rejected; appointment data remains consistent.
Actual Result	[To be recorded during execution]
Status	Not Run

TC-COL-01 — Blood collection and donor update

Requirement ID	BBMS-F-017–F-019
Module / Feature	Blood Collection
Priority	High
Objective	Verify staff can record collection details and donor history is updated immediately.
Preconditions	Confirmed donor appointment and staff account.
Test Data	Collection volume and timestamp.
Test Steps	1. Start collection. 2. Record volume/date/time. 3. Complete collection. 4. Check donor history and last-donation date.
Expected Result	Collection is stored against appointment; required volume/time are captured; donor last-donation date and history update immediately.
Actual Result	[To be recorded during execution]
Status	Not Run

TC-BAG-01 — Bag ID uniqueness

Requirement ID	BBMS-F-020, F-021
Module / Feature	Blood Unit
Priority	High
Objective	Verify a unique Bag ID is generated and duplicates are rejected.
Preconditions	Collection can be completed.
Test Data	Two collection events; forced duplicate Bag ID attempt.
Test Steps	1. Complete first collection. 2. Record generated Bag ID. 3. Complete second collection. 4. Attempt to reuse the first Bag ID.
Expected Result	Each collection receives a unique Bag ID; duplicate Bag ID creation is rejected.
Actual Result	[To be recorded during execution]
Status	Not Run

TC-BAG-02 — Blood group management

Requirement ID	BBMS-F-022–F-023
Module / Feature	Blood Group Management
Priority	High
Objective	Verify all eight ABO/Rh groups are supported and staff can store the lab-confirmed group separately from donor self-report.
Preconditions	Staff account and blood unit records available.
Test Data	A+, A-, B+, B-, AB+, AB-, O+, O-; differing donor self-report and lab-confirmed result.
Test Steps	1. Enter each supported group. 2. Verify persistence. 3. Record a lab-confirmed correction distinct from donor value.
Expected Result	All eight groups are accepted and retained; confirmed blood group is stored independently.
Actual Result	[To be recorded during execution]
Status	Not Run

TC-SER-01 — Serology quarantine and availability

Requirement ID	BBMS-F-024–F-027
Module / Feature	Serology / Quarantine
Priority	High
Objective	Verify all required tests are recorded and unsafe units cannot become available.
Preconditions	Collected unit exists in Quarantined state.
Test Data	Pass results for all tests; separate case with one Fail.
Test Steps	1. Record incomplete tests. 2. Verify unit remains quarantined. 3. Record all Pass results. 4. Verify status becomes Available. 5. Record a Fail for another unit.
Expected Result	Unit remains quarantined until all tests pass; all-pass unit can become Available; any Fail sets Discarded and blocks allocation/dispatch.
Actual Result	[To be recorded during execution]
Status	Not Run

TC-SER-02 — Serology result recording completeness

Requirement ID	BBMS-F-024–F-025
Module / Feature	Serology Result Entry
Priority	High
Objective	Verify each required serology test accepts Pass/Fail and incomplete test sets cannot make a unit Available.
Preconditions	Collected unit in Quarantined state.
Test Data	HIV, Hepatitis B, Hepatitis C, Syphilis results including incomplete and mixed outcomes.
Test Steps	1. Enter an incomplete result set. 2. Verify status. 3. Enter all Pass results. 4. Repeat with one Fail.
Expected Result	Incomplete set leaves unit Quarantined; all Pass supports Available; any Fail supports Discarded.
Actual Result	[To be recorded during execution]
Status	Not Run

TC-QNT-01 — Quarantine access and status audit

Requirement ID	BBMS-F-028–F-030
Module / Feature	Quarantine
Priority	High
Objective	Verify unsafe units are excluded from availability/search, quarantine records are auditable, and authorized staff can view quarantine duration.
Preconditions	Quarantined and discarded units exist; staff account available.
Test Data	Units with quarantine transitions.
Test Steps	1. Search inventory/public data. 2. Verify excluded statuses. 3. Open staff quarantine list. 4. Change quarantine status. 5. Inspect audit entry.
Expected Result	Unsafe units are excluded; authorized staff sees current quarantine duration; status transition logs actor and timestamp.
Actual Result	[To be recorded during execution]
Status	Not Run

TC-CMP-01 — Component separation traceability

Requirement ID	BBMS-F-031–F-032
Module / Feature	Component Separation
Priority	Medium
Objective	Verify a whole-blood unit can be separated into distinct components and each component retains the source Bag ID.
Preconditions	Eligible whole-blood unit available to staff.
Test Data	One whole-blood unit and component set: RBC/Plasma/Platelets.
Test Steps	1. Record separation. 2. Review component inventory records. 3. Verify source Bag ID reference.
Expected Result	Each resulting component has a distinct inventory record linked to the originating Bag ID.
Actual Result	[To be recorded during execution]
Status	Not Run

TC-INV-01 — Inventory counts and FEFO

Requirement ID	BBMS-F-033–F-039
Module / Feature	Inventory / FEFO
Priority	High
Objective	Verify count updates, expiry handling, and first-expiring selection.
Preconditions	At least two available compatible units with different expiry dates.
Test Data	Units U1/U2 with ordered expiry dates; one expired unit.
Test Steps	1. Verify counts. 2. Allocate a unit. 3. Recheck count. 4. Run allocation suggestion. 5. Advance/test expiry condition.
Expected Result	Counts update correctly; FEFO selects the closest expiry; expired units are flagged and excluded from counts/allocation.
Actual Result	[To be recorded during execution]
Status	Not Run

TC-INV-02 — Inventory availability count updates

Requirement ID	BBMS-F-033–F-036
Module / Feature	Inventory Counts
Priority	High
Objective	Verify inventory counts increase when units become Available and decrease on allocation.
Preconditions	Inventory dashboard available; test units in controlled states.
Test Data	Available and quarantined units across groups/components.
Test Steps	1. Record baseline count. 2. Mark a passed unit Available. 3. Verify count increment. 4. Allocate a unit. 5. Verify decrement.
Expected Result	Counts reflect only available units and update correctly after state changes.
Actual Result	[To be recorded during execution]
Status	Not Run

TC-EXP-01 — Expiry handling

Requirement ID	BBMS-F-037–F-039
Module / Feature	Expiry / FEFO
Priority	High
Objective	Verify component-specific expiry dates are calculated, expired units are flagged, and expired units are excluded from allocation.
Preconditions	Available units with different component types and controlled test dates.
Test Data	RBC/whole blood, Platelets, frozen Plasma; one expired unit.
Test Steps	1. Mark units Available. 2. Verify expiry calculation. 3. Trigger/advance expiry. 4. Attempt allocation of expired unit.
Expected Result	Expiry dates follow configured component rules; expired unit is flagged and cannot be counted or allocated.
Actual Result	[To be recorded during execution]
Status	Not Run

TC-AUTH-03 — Hospital registration, approval and login

Requirement ID	BBMS-F-040–F-042
Module / Feature	Hospital Registration / Login
Priority	High
Objective	Verify hospital registration requires administrator approval and unapproved hospitals cannot submit requisitions.
Preconditions	Administrator and hospital test accounts available.
Test Data	Valid hospital registration data; pending and approved states.
Test Steps	1. Submit hospital registration. 2. Attempt login/requisition while pending. 3. Approve as administrator. 4. Log in and submit a test request.
Expected Result	Pending hospital cannot use requisition functions; after approval, authentication and requisition access are enabled.
Actual Result	[To be recorded during execution]
Status	Not Run

TC-REQ-01 — Routine blood requisition lifecycle

Requirement ID	BBMS-F-043–F-046
Module / Feature	Hospital Requisition
Priority	High
Objective	Verify valid requisition creation, unique ID, staff decision and hospital tracking.
Preconditions	Approved hospital user and available staff account.
Test Data	Blood group, component, quantity and required-by date.
Test Steps	1. Submit requisition. 2. Confirm unique ID and Pending status. 3. Approve/partially fulfil/reject as a staff user. 4. Review hospital history.
Expected Result	Required requisition fields are accepted; unique ID and Pending status are set; staff can approve/partially fulfil/reject and rejection reason is recorded; hospital sees current status/history.
Actual Result	[To be recorded during execution]
Status	Not Run

TC-REQ-02 — Requisition decision and tracking

Requirement ID	BBMS-F-043–F-046
Module / Feature	Hospital Requisition
Priority	High
Objective	Verify staff can approve, partially fulfil, or reject a requisition and the hospital sees status history.
Preconditions	Approved hospital user and pending requisition.
Test Data	Routine requisition with enough and insufficient inventory; rejection reason.
Test Steps	1. Submit request. 2. Approve one case. 3. Partially fulfil another. 4. Reject a third with reason. 5. View hospital history.
Expected Result	Each staff action updates the correct status and rejection reason; hospital history reflects current and prior statuses.
Actual Result	[To be recorded during execution]
Status	Not Run

TC-REQ-03 — Emergency/STAT prioritization

Requirement ID	BBMS-F-047, F-048
Module / Feature	Hospital Requisition
Priority	High
Objective	Verify Emergency/STAT requests are prioritized and staff are notified.
Preconditions	Approved hospital user and pending routine request.
Test Data	Routine request and later Emergency/STAT request.
Test Steps	1. Submit routine request. 2. Submit Emergency/STAT request. 3. Open staff queue. 4. Check alert/notification.
Expected Result	STAT request appears at top of pending queue and available staff receive an on-screen alert and/or email immediately.
Actual Result	[To be recorded during execution]
Status	Not Run

TC-MATCH-01 — ABO/Rh compatibility and allocation restriction

Requirement ID	BBMS-F-049–F-051
Module / Feature	Cross-Matching
Priority	High
Objective	Verify compatible units are accepted and incompatible units are blocked unless authorized override is recorded.
Preconditions	Approved requisition and inventory containing compatible/incompatible units.
Test Data	Compatible and incompatible blood groups; optional override justification.
Test Steps	1. View compatible group list. 2. Attempt incompatible allocation. 3. Attempt authorized override with justification. 4. Review recorded outcome.
Expected Result	Compatible groups are displayed; incompatible allocation is prevented unless authorized override is recorded; outcome is logged as Compatible/Incompatible/Overridden.
Actual Result	[To be recorded during execution]
Status	Not Run

TC-DISP-01 — Allocation and dispatch

Requirement ID	BBMS-F-052–F-055
Module / Feature	Dispatch
Priority	High
Objective	Verify allocation by Bag ID, dispatch record, status changes, and fulfilment status.
Preconditions	Approved requisition with sufficient compatible inventory.
Test Data	One or more Bag IDs.
Test Steps	1. Allocate units to requisition. 2. Generate dispatch record. 3. Dispatch units. 4. Review inventory and requisition status.
Expected Result	Allocated Bag IDs are linked to requisition; dispatch record captures requisition, Bag IDs, staff and timestamp; units become Dispatched and are removed from available counts; requisition becomes Fulfilled or Partially Fulfilled as applicable.
Actual Result	[To be recorded during execution]
Status	Not Run

TC-DEL-01 — Delivery confirmation

Requirement ID	BBMS-F-056, F-057
Module / Feature	Delivery
Priority	Medium
Objective	Verify hospital receipt confirmation updates custody records and unit status.
Preconditions	Dispatched unit linked to hospital requisition.
Test Data	Receiving hospital user.
Test Steps	1. Open delivered requisition. 2. Confirm receipt. 3. Review receipt record and unit status.
Expected Result	Timestamp and receiving staff identity are recorded; unit status changes to Delivered.
Actual Result	[To be recorded during execution]
Status	Not Run

TC-AUD-01 — Audit logging and read-only access

Requirement ID	BBMS-F-058, F-059
Module / Feature	Audit
Priority	High
Objective	Verify data-changing actions create audit records and only administrators can view/edit audit entries.
Preconditions	Users in multiple roles; test records available.
Test Data	Create/update/delete actions and role changes.
Test Steps	1. Perform representative data changes. 2. Review audit log as Administrator. 3. Attempt audit access/edit as non-admin.
Expected Result	Audit entries contain actor, action, timestamp and record ID; non-admin access is denied; audit entries are read-only and non-deletable.
Actual Result	[To be recorded during execution]
Status	Not Run

TC-ADM-01 — Administrator user and role management

Requirement ID	BBMS-F-060–F-063
Module / Feature	Admin / User / Role Management
Priority	High
Objective	Verify administrators can manage accounts/roles, approve hospitals, and deactivated users cannot authenticate.
Preconditions	Administrator account and test accounts for all roles.
Test Data	New user; role change; hospital registration; deactivate/reactivate operations.
Test Steps	1. Create user. 2. Assign/review role. 3. Approve/reject hospital. 4. Deactivate account and attempt login. 5. Reactivate and retry.
Expected Result	Admin actions follow permissions; deactivated account is denied; reactivated account can authenticate again.
Actual Result	[To be recorded during execution]
Status	Not Run

TC-RPT-01 — Reports and analytics

Requirement ID	BBMS-F-064–F-066
Module / Feature	Reports
Priority	Medium
Objective	Verify required report data and date-range filtering.
Preconditions	Inventory, donation and requisition test records exist.
Test Data	Selected date range and records.
Test Steps	1. Generate inventory report. 2. Generate donation activity report. 3. Generate requisition fulfilment report. 4. Compare totals with underlying test records.
Expected Result	CSV inventory report contains group/component levels; donation report shows totals/units collected/discarded; fulfilment report shows received/fulfilled/partial/rejected counts for selected date range.
Actual Result	[To be recorded during execution]
Status	Not Run

TC-PUB-01 — Public blood availability and privacy

Requirement ID	BBMS-F-067–F-068
Module / Feature	Public Portal
Priority	High
Objective	Verify unauthenticated availability lookup and absence of identifying information.
Preconditions	Public portal available with synthetic stock.
Test Data	Inventory across blood groups/locations; donor/hospital identifying records.
Test Steps	1. Open public page without login. 2. Verify current availability by group/location. 3. Inspect page/API payload for identifiers.
Expected Result	Availability is visible without authentication and no donor-identifying or hospital-identifying information is exposed.
Actual Result	[To be recorded during execution]
Status	Not Run

TC-NF-01 — Performance under concurrent load

Requirement ID	BBMS-NF-001–NF-003
Module / Feature	Performance
Priority	High
Objective	Measure blood-availability search, appointment booking and authenticated-user response limits.
Preconditions	Stable test build; load tool configured.
Test Data	Synthetic dataset and up to 50 concurrent sessions.
Test Steps	1. Run blood-availability search load. 2. Run appointment-booking transaction timings. 3. Run concurrent authenticated-user scenario. 4. Record response metrics.
Expected Result	Availability search completes within 2 seconds under up to 50 concurrent users; booking completes within 3 seconds; concurrency requirement is met without exceeding stated response limit.
Actual Result	[To be recorded during execution]
Status	Not Run

TC-NF-02 — Reliability, restart persistence and error logging

Requirement ID	BBMS-NF-004–NF-006
Module / Feature	Reliability
Priority	High
Objective	Verify committed transactions survive application restart and unhandled errors are logged while availability target is assessed.
Preconditions	Test build with persistence enabled.
Test Data	Committed donation, requisition and dispatch records; controlled fault case.
Test Steps	1. Create and commit records. 2. Restart application. 3. Verify records. 4. Trigger controlled unhandled error path. 5. Review error log.
Expected Result	Committed transaction data persists without loss; unhandled errors are logged with timestamp and stack trace; availability evidence is collected for scheduled test period.
Actual Result	[To be recorded during execution]
Status	Not Run

TC-NF-03 — Usability and accessibility validation

Requirement ID	BBMS-NF-007–NF-009
Module / Feature	Usability / Accessibility
Priority	Medium
Objective	Verify registration duration, WCAG 2.1 AA requirements, and error-message clarity.
Preconditions	Accessible browser build and primary workflows available.
Test Data	First-time user scenario; invalid form entries.
Test Steps	1. Time first-time registration. 2. Run Lighthouse/manual accessibility checks. 3. Submit invalid form values.
Expected Result	Registration completes in under 5 minutes without assistance; primary workflows satisfy the specified WCAG color-contrast and keyboard-navigation criteria; errors clearly identify field and issue.
Actual Result	[To be recorded during execution]
Status	Not Run

TC-NF-04 — Maintainability and unit-test coverage review

Requirement ID	BBMS-NF-010–NF-011
Module / Feature	Maintainability
Priority	Low
Objective	Verify modular/layered separation and minimum backend unit-test coverage.
Preconditions	Source tree and test suite available.
Test Data	Backend repository and coverage report.
Test Steps	1. Review module boundaries. 2. Verify inventory change does not require unrelated reporting changes. 3. Generate unit-test coverage report.
Expected Result	Backend follows layered/modular structure and backend business-logic coverage is at least 60%.
Actual Result	[To be recorded during execution]
Status	Not Run

TC-NF-05 — Browser and runtime portability

Requirement ID	BBMS-NF-012–NF-013
Module / Feature	Portability
Priority	Medium
Objective	Verify client rendering on required browsers and backend deployment on supported OS/runtime.
Preconditions	Access to required browser versions and one supported host.
Test Data	Primary BBMS workflows.
Test Steps	1. Execute smoke workflow on latest two stable Chrome/Firefox/Edge. 2. Deploy backend on supported Python/Node runtime host. 3. Execute smoke tests.
Expected Result	Web client renders correctly on required browser versions; backend deploys without code modification on supported runtime/host.
Actual Result	[To be recorded during execution]
Status	Not Run

TC-NF-06 — Scalability architecture review

Requirement ID	BBMS-NF-014–NF-015
Module / Feature	Scalability
Priority	Low
Objective	Verify data-access abstraction supports future RDBMS migration and new blood-bank locations.
Preconditions	Source, schema, and architecture artifacts available.
Test Data	Database schema and location-related modules.
Test Steps	1. Inspect data-access interfaces. 2. Review location model boundaries. 3. Verify no donor/hospital/inventory module changes are inherently required for a new location.
Expected Result	SQLite-specific persistence is isolated behind data-access interfaces and the architecture supports a new location without modifying donor, hospital, or inventory modules as specified.
Actual Result	[To be recorded during execution]
Status	Not Run

TC-SEC-01 — Authentication enforcement

Requirement ID	BBMS-SR-001
Module / Feature	Security / Authentication
Priority	High
Objective	Verify unauthenticated users cannot access role-specific functions while valid authenticated users can access permitted functions.
Preconditions	Protected endpoint and valid test accounts.
Test Data	Unauthenticated request; valid donor, hospital, staff, and admin sessions.
Test Steps	1. Access protected endpoint without login. 2. Repeat with valid sessions for each role. 3. Verify role-specific access.
Expected Result	Unauthenticated access is denied; authenticated users are admitted only to authorized functions.
Actual Result	[To be recorded during execution]
Status	Not Run

TC-SEC-02 — Password hashing and plaintext protection

Requirement ID	BBMS-SR-002
Module / Feature	Security / Authentication
Priority	High
Objective	Verify passwords are stored as salted one-way hashes and are not exposed.
Preconditions	Controlled test account and database access for inspection.
Test Data	Synthetic password.
Test Steps	1. Register/change password. 2. Inspect credential storage. 3. Inspect server logs and API responses.
Expected Result	Only salted one-way hash representation is stored; plaintext password is absent from logs and responses.
Actual Result	[To be recorded during execution]
Status	Not Run

TC-SEC-03 — Server-side RBAC enforcement

Requirement ID	BBMS-SR-003
Module / Feature	Security / Authorization
Priority	High
Objective	Verify each protected API endpoint enforces server-side role authorization.
Preconditions	Donor, Hospital, Staff and Administrator accounts.
Test Data	Unauthorized endpoint requests.
Test Steps	1. Authenticate as each lower-privilege role. 2. Call protected endpoints directly. 3. Record HTTP responses.
Expected Result	Unauthorized requests are rejected with HTTP 403 and no protected operation/data is exposed.
Actual Result	[To be recorded during execution]
Status	Not Run

TC-SEC-04 — SQL injection prevention

Requirement ID	BBMS-SR-004
Module / Feature	Security / Database
Priority	High
Objective	Verify raw SQL injection attempts cannot alter query semantics or access unauthorized data.
Preconditions	Test environment with synthetic database.
Test Data	Common SQL injection payloads in authentication and search inputs.
Test Steps	1. Submit payloads. 2. Observe API responses and server behavior. 3. Verify no unintended data access or state change.
Expected Result	Payloads are treated as data; database operations remain parameterized/ORM-backed; no SQL injection effect occurs.
Actual Result	[To be recorded during execution]
Status	Not Run

TC-SEC-05 — Input validation and XSS prevention

Requirement ID	BBMS-SR-005
Module / Feature	Security / Input
Priority	High
Objective	Verify malformed inputs are rejected and stored/reflected XSS is prevented.
Preconditions	Forms and API endpoints available.
Test Data	Oversized values, invalid types, script/event-handler payloads.
Test Steps	1. Submit malformed payloads. 2. Submit XSS probes in stored and reflected contexts. 3. Reload affected pages.
Expected Result	Invalid input is rejected/validated; output is safely encoded; scripts do not execute.
Actual Result	[To be recorded during execution]
Status	Not Run

TC-SEC-06 — Session and cookie security

Requirement ID	BBMS-SR-006
Module / Feature	Security / Session
Priority	High
Objective	Verify cookie flags, logout invalidation and inactivity timeout.
Preconditions	Authenticated browser session.
Test Data	Session cookie and controlled inactivity period.
Test Steps	1. Inspect cookie attributes. 2. Logout and retry protected request. 3. Simulate 30 minutes inactivity.
Expected Result	Session cookies are HttpOnly/Secure with defined expiry; logout invalidates the session; inactivity beyond 30 minutes invalidates the session.
Actual Result	[To be recorded during execution]
Status	Not Run

TC-SEC-07 — Brute-force lockout and rate limiting

Requirement ID	BBMS-SR-007
Module / Feature	Security / Authentication
Priority	High
Objective	Verify repeated failed attempts trigger lockout and rapid requests are rate-limited.
Preconditions	Controlled test account and client IP.
Test Data	Six failed attempts; >10 login requests within one minute.
Test Steps	1. Submit five consecutive failed login attempts. 2. Attempt sixth login during lockout. 3. Generate rapid requests within one minute.
Expected Result	Account is locked for 15 minutes after five failures; rapid login traffic beyond 10 requests per minute per source IP receives rate limiting.
Actual Result	[To be recorded during execution]
Status	Not Run

TC-SEC-08 — HTTPS/TLS and security-event auditing

Requirement ID	BBMS-SR-008
Module / Feature	Security / Transport / Audit
Priority	High
Objective	Verify encrypted communication and tamper-evident logging of security-relevant events.
Preconditions	HTTPS-enabled test environment.
Test Data	Login attempts, lockout and role-change actions.
Test Steps	1. Access application over HTTP and HTTPS. 2. Perform security-relevant events. 3. Review security audit records.
Expected Result	Client-server communication uses HTTPS/TLS; security-relevant events such as login attempts, lockouts and role changes are recorded in the audit trail.
Actual Result	[To be recorded during execution]
Status	Not Run

14. Test Metrics & Reporting

Metrics collected:

- % of test cases executed (of the 41 planned cases).
- % passed / failed / blocked.
- Defect density (defects per module) and defect aging (days from log to closure).
- Requirement coverage (% of SRS requirement groups in the RTM with at least one executed test).
- Performance measurements for NF-001 to NF-003 (median, 95th percentile and maximum response time under load).

- Backend unit-test coverage (target $\geq 60\%$, NF-011).

Reports:

- Test execution status report at each milestone (T2–T5) and on request.
- Defect summary with severity and aging.
- Final Test Summary Report with the completed RTM (milestone T6).

15. Approvals

Approval is to be completed by the designated project/team authority and course evaluator after review. No approval is implied by this draft document.

Role	Name	Signature / Date
Project / Test Lead		
Faculty / Course Evaluator		